

Cost-Effectiveness of Intismeran Autogene Plus Pembrolizumab Versus Pembrolizumab Alone as Adjuvant Therapy for Resected Stage IIIB–IV Melanoma: An Updated Early Economic Evaluation

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Abstract

Background: In August 2026, Merck and Moderna reported positive Phase 3 INTerpath-001 results for intismeran autogene (mRNA-4157/V940) plus pembrolizumab in resected stage IIB–IV melanoma. Its cost-effectiveness remains unknown.

Objective: To estimate the cost-effectiveness of intismeran plus pembrolizumab versus pembrolizumab alone from a US payer perspective.

Methods: A four-state partitioned survival model was developed with a 40-year horizon and 3% annual discounting. Survival functions were calibrated to the 5-year KEYNOTE-942 trial follow-up. Health-state utilities were drawn from published EQ-5D literature. Probabilistic, deterministic, and scenario sensitivity analyses were conducted.

Results: In the base case, intismeran plus pembrolizumab yielded 14.16 QALYs at a cost of \$1,070,705 versus 7.65 QALYs at \$745,939 for pembrolizumab alone. The ICER was \$49,841/QALY (incremental cost \$324,765). Probabilities of cost-effectiveness were 93.4% at \$100,000 and 98.4% at \$150,000/QALY. Deterministic threshold prices were \$527K and \$852K. Results were robust across scenarios; the ICER remained below \$100,000/QALY except for hazard-convergence waning (\$104,247/QALY).

Conclusion: At an estimated acquisition price of \$200,000 per course, intismeran plus pembrolizumab was projected to be cost-effective under conventional US thresholds, driven by a 6.52-QALY gain despite higher downstream disease-management costs.

Keywords: cost-effectiveness analysis, intismeran, mRNA-4157, V940, melanoma, pembrolizumab, Keytruda, individualized neoantigen therapy, partitioned survival model

1 Introduction

1.1 A New Reimbursement Problem

On August 19, 2026, Merck and Moderna announced positive topline results from the Phase 3 INTERpath-001 trial, establishing intismeran autogene (mRNA-4157/V940) plus pembrolizumab as the first mRNA-based individualized neoantigen therapy (INT) to succeed in a late-stage trial [2]. The press release confirmed statistically significant improvements in RFS and DMFS, though numerical hazard ratios have not yet been disclosed. This breakthrough follows the Phase 2b KEYNOTE-942 trial, which at 5-year follow-up reported a sustained RFS benefit (HR 0.51, 95% CI 0.294–0.887), DMFS benefit (HR 0.411, 95% CI 0.200–0.843), and 5-year OS rates of 92.2% (combo) versus 71.3% (pembrolizumab alone; OS HR 0.471, 95% CI 0.165–1.345) [3].

Despite these clinical advances, the economic implications of INTs remain poorly understood. These therapies combine the high costs of immune checkpoint inhibitors with the additional expense of individualized manufacturing, creating a reimbursement challenge unlike any in oncology.

1.2 Why INTs Differ from Conventional Oncology Pricing

Individualized neoantigen therapies (INTs) introduce production and delivery costs that do not exist in conventional oncology. Unlike fixed-composition drugs, each INT treatment is custom-manufactured for a single patient based on tumor sequencing data, requiring mRNA synthesis, lipid nanoparticle formulation, and quality assurance for each batch [4, 5]. These costs scale with each new patient rather than benefiting from traditional manufacturing economies of scale.

Jefferies analysts have estimated the cost of such vaccines at \$100,000–\$300,000 per patient [6]. By comparison, cancer immunotherapies such as Keytruda® (pembrolizumab) have annual costs of approximately \$220,896 in the adjuvant setting [7]. The combination of a high-cost immunotherapy with an individualized vaccine creates a compound affordability challenge for payers, particularly in the adjuvant setting where patients are recurrence-free at the time of treatment.

1.3 Existing Economic Evidence

A prior exploratory cost-effectiveness analysis of V940 plus pembrolizumab in adjuvant melanoma was presented at ISPOR 2024 [8]. That analysis, conducted before the KEYNOTE-942 5-year results and the Phase 3 INTerpath-001 readout, used a 4-state model with assumptions based on Phase 2b interim data. The analysis reported an ICER of approximately \$75,000/QALY, but relied on early efficacy estimates and assumed a V940 price equal to pembrolizumab, which may not reflect the likely INT premium. The present analysis updates these findings by incorporating the mature 5-year KEYNOTE-942 data, applying a ratio-constrained partition survival structure with a general-population hazard floor, and conducting a comprehensive deterministic threshold price analysis.

1.4 Clinical Basis and Research Objectives

The clinical foundation for this analysis comes from the 5-year follow-up of the Phase 2b KEYNOTE-942 trial, which provides the most mature efficacy data available for intismeran [3]. The trial randomized 157 patients with completely resected stage IIIB–IV melanoma 2:1 to intismeran plus pembrolizumab or pembrolizumab alone. The primary endpoint was RFS; secondary endpoints included DMFS, OS, and safety. At the 5-year analysis, the combination showed a sustained RFS benefit with a hazard ratio of 0.51 (95% CI 0.294–0.887), a DMFS HR of 0.411 (95% CI 0.200–0.843), and OS data still maturing with 5-year OS rates of 92.2% (combo) versus 71.3% (pembrolizumab alone; OS HR 0.471, 95% CI 0.165–1.345).

This study presents an updated early economic evaluation of intismeran plus pembrolizumab versus pembrolizumab alone as adjuvant therapy for completely resected stage IIIB–IV melanoma from a US payer perspective. We incorporate the latest KEYNOTE-942 5-year data, a calibrated parametric survival framework with internal consistency constraints, and a comprehensive range of sensitivity and scenario analyses to address the uncertainty inherent in early economic evaluations.

2 Methods

2.1 Model Structure

A partitioned survival model (PSM) was developed in Python with four mutually exclusive health states: recurrence-free (RF), locoregional recurrence (LR), distant metastasis (DM), and death, following the structure of prior melanoma cost-effectiveness models and standard PSM methodology [9–12]. Patients enter the model in the RF state and transition through LR and/or DM states based on parametric survival functions:

$$\text{RF}(t) = \text{RFS}(t)$$

$$\text{LR}(t) = \text{DMFS}(t) - \text{RFS}(t)$$

$$\text{DM}(t) = \text{OS}(t) - \text{DMFS}(t)$$

$$\text{Dead}(t) = 1 - \text{OS}(t)$$

This construction ensures $\text{RFS} \leq \text{DMFS} \leq \text{OS}$ at all times and maintains internal consistency without forcing a specific parametric form on the transition hazards.

2.2 Clinical Data

RFS and OS survival probabilities were extracted at 18, 24, 36, 48, and 60 months, and DMFS probabilities at 18, 24, 36, and 48 months, from the published KEYNOTE-942 Kaplan–Meier curves [3] (60-month DMFS was not reported in the 5-year analysis). Parametric survival models were fitted to these data points using weighted nonlinear least squares, following standard methodological guidance for survival extrapolation in health economic evaluation [13–15]. Model selection was performed by comparing five candidate distributions (exponential, Weibull, log-normal, log-logistic, generalized gamma) for each of the three underlying survival components (OS, $r_{\text{DM}} = \text{DMFS}/\text{OS}$, $r_{\text{LR}} = \text{RFS}/\text{DMFS}$) using Akaike Information Criterion (AIC), as recommended by NICE DSU TSD 14 [13]. The log-normal distribution was selected as the base case because it achieved the best AIC for the OS curve of the combination arm (the most influential curve driving lifetime costs and QALYs) and was broadly competitive across

all other components (Supplementary Table S3). All 4,096 candidate distribution combinations preserved the structural ordering constraint $RFS \leq DMFS \leq OS$ at 40 years, so the choice of distribution was not driven by feasibility constraints. A common log-normal family was chosen for parsimony and interpretability; component-specific best-fitting distributions (Weibull, log-logistic) were examined in scenario analyses. The fitted model was calibrated to ensure the constraint $RFS \leq DMFS \leq OS$ was satisfied at all time points. To address the external validity of long-term survival extrapolation, a general population mortality hazard floor was applied from 60 months onward, using the age- and sex-matched US 2019 life table (pooled 64% male/36% female, baseline age 61 years) [16]. This constraint ensures that the model does not produce survival probabilities exceeding those of the general population at any time point, a standard practice in oncology cost-effectiveness analyses [10, 13].

2.3 External Validation

Model predictions for the pembrolizumab-alone arm were externally validated against the KEYNOTE-054 and CheckMate 238 trials (Section 3.3). Figure 1 illustrates the fitted RFS, DMFS, and OS trajectories for both treatment arms together with the observed KEYNOTE-942 landmark data used for calibration.

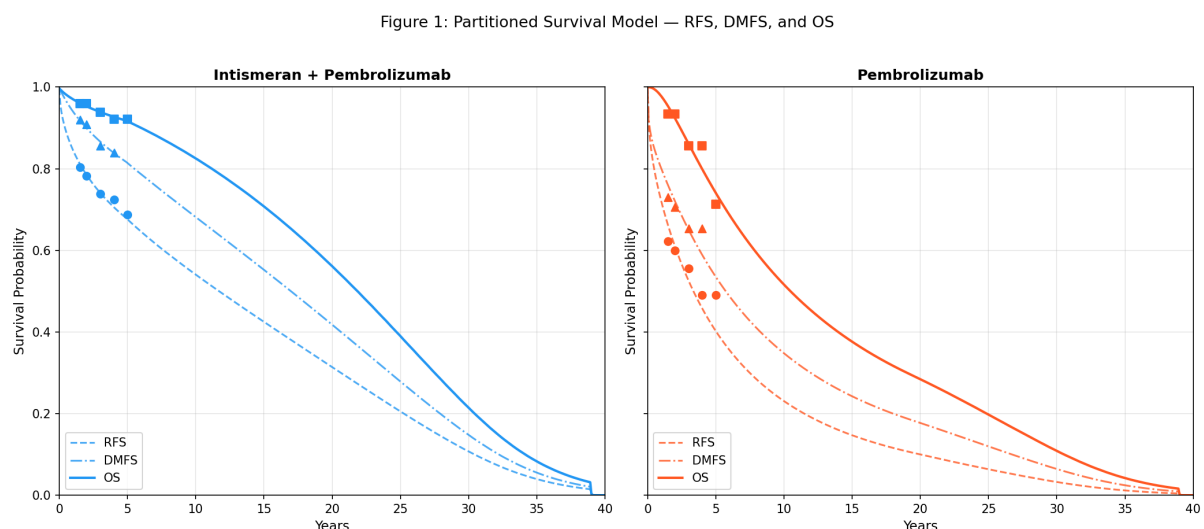


Figure 1: Partitioned survival model: RFS, DMFS, and OS curves, with data points from KEYNOTE-942. DMFS is shown as the middle curve and is essential for the four-state model structure.

2.4 Costs

The analysis adopted a US payer perspective. Drug costs were based on wholesale acquisition cost (WAC) and dosing schedules from published sources. Keytruda® (pembrolizumab) was assumed at 200 mg every 3 weeks for up to 18 cycles at a WAC of \$12,272 per 200 mg dose, yielding an annual cost of \$220,896 [7]. Intismeran was modeled at an assumed course acquisition price of \$200,000 based on Jefferies analyst estimates [6].

Administration costs were included for intravenous infusions (\$500 per cycle, based on the CMS Physician Fee Schedule [17]). Adverse event costs were incorporated at \$15,000 incremental for the combination arm, based on published melanoma cost-effectiveness analyses [10, 11]. In the KEYNOTE-942 5-year safety analysis, treatment-related Grade 3–5 adverse events occurred in 25.0% of combination-arm patients and 14.0% of pembrolizumab-arm patients (Supplementary Table A4, [3]). Monthly disease management costs were \$3,000 for the locoregional recurrence state (salvage surgery and adjuvant therapy amortized) and \$12,000 for the distant metastasis state (systemic therapy for advanced melanoma), consistent with published melanoma CEA literature [10, 11]. All costs were inflated to 2026 USD using the medical care component of the Consumer Price Index [18].

Table 1: Model Input Parameters

Parameter	Base Case	Source
5-year RFS, combo	68.8%	KEYNOTE-942 [3]
5-year RFS, pembro	49.1%	KEYNOTE-942 [3]
RFS HR	0.51 (95% CI 0.294–0.887)	KEYNOTE-942 [3]
DMFS HR	0.411 (95% CI 0.200–0.843)	KEYNOTE-942 [3]
OS HR	0.471 (95% CI 0.165–1.345)	KEYNOTE-942 [3]
5-year OS, combo	92.2%	KEYNOTE-942 [3]
5-year OS, pembro	71.3%	KEYNOTE-942 [3]
Keytruda annual cost	\$220,896	WAC [7]
Intismeran course cost	\$200,000	Jefferies [6]

2.5 Utilities

Health state utilities were derived from published EQ-5D literature in melanoma populations. The base case utilities were: RF 0.83, LR 0.64, and DM 0.55 [10, 11, 19]. Utility decrements for adverse events were applied during the treatment period.

2.6 Base Case Analysis

The base case analysis was conducted over a lifetime horizon (40 years) with 3% annual discounting for both costs and QALYs, following the recommendations of the Second Panel on Cost-Effectiveness in Health and Medicine [20]. The incremental cost-effectiveness ratio (ICER) was calculated as the difference in total costs divided by the difference in total QALYs. Cost-effectiveness was evaluated at willingness-to-pay thresholds of \$50,000, \$100,000, and \$150,000/QALY [21].

2.7 Sensitivity Analysis

Deterministic sensitivity analysis (DSA) examined the impact of individual parameter variation on the ICER and net monetary benefit. Parameters were varied one at a time across plausible ranges: survival model parameters, health-state utilities, costs, and the discount rate (Table 1).

Probabilistic sensitivity analysis (PSA) was conducted with 6,000 Monte Carlo iterations to propagate joint parameter uncertainty [22]. Survival parameters (μ, σ) for all six underlying curves (OS, r_{DM} , r_{LR} , each for the combination and pembrolizumab arms) were jointly sampled from bivariate normal distributions with covariance matrices derived from the non-linear least-squares Jacobian (Supplementary Table S12). Cross-arm covariance could not be estimated from the published aggregate Kaplan–Meier data; arm-specific survival parameters were therefore sampled independently. No specific cross-arm correlation was imposed because such a parameter could not be identified without patient-level data. Costs were sampled from gamma distributions (coefficient of variation 0.2), and utilities from beta distributions (standard error 0.03). The intismeran acquisition price was held fixed at \$200,000 in all iterations. Structural constraints ($RFS \leq DMFS \leq OS$, monotonicity, survival within [0,1]) were verified in

every iteration; no iterations were excluded on the basis of the ICER value, and all 6,000 draws satisfied the structural validity criteria (0 rejections). Results are reported as mean incremental net monetary benefit (NMB) and cost-effectiveness probability at each willingness-to-pay threshold, following recommended practice [23, 24]. Convergence was verified by checking that cumulative means stabilised by approximately 500 iterations (Supplementary Figure S6).

3 Results

3.1 Base-Case Clinical and Economic Outcomes

In the base case with general population mortality constraint, intismeran plus pembrolizumab generated an additional 8.26 life-years (19.01 vs 10.75 life-years) and 6.52 additional QALYs (14.16 vs 7.65 QALYs) compared with pembrolizumab alone, at an incremental lifetime cost of \$324,765 (\$1,070,705 vs \$745,939). This yielded an ICER of \$49,841/QALY gained (Table 2). At a willingness-to-pay threshold of \$100,000/QALY, the incremental net monetary benefit was \$326,840; at \$150,000/QALY, it was \$652,643 (Table 2).

NMB, net monetary benefit ($=\lambda \times \text{QALYs} - \text{cost}$, where λ is the willingness-to-pay threshold) [23].

3.2 Cost and QALY Decomposition

The \$200,000 acquisition cost of intismeran contributed substantially to the incremental cost, but the total incremental cost of \$324,765 also reflects higher lifetime disease-management expenditure in the combination arm, because combination-arm patients survive substantially longer and thus accumulate more time in the LR and DM health states (Table 2). Specifically, the combination arm had \$37,726 higher locoregional-recurrence management costs (\$112,203 vs \$74,477) and \$71,039 higher distant-metastasis management costs (\$524,121 vs \$453,082) than the pembrolizumab-alone arm, reflecting the longer overall survival and correspondingly greater cumulative exposure to disease-management costs. The incremental QALY gain of 6.52 was driven almost entirely by additional time spent in the recurrence-free state (5.57 RF QALYs gained), which more than compensated for modestly more time in the DM state (0.27

DM QALYs).

Table 2: Base-Case Results and Decomposition

Component	Combo	Pembro	Incremental
<i>Cost decomposition</i>			
Intismeran	\$200,000	\$0	\$200,000
Pembrolizumab	\$217,888	\$217,888	\$0
Sequencing	\$1,000	\$0	\$1,000
Administration/AE	\$15,493	\$493	\$15,000
LR management	\$112,203	\$74,477	\$37,726
DM management	\$524,121	\$453,082	\$71,039
Total cost	\$1,070,705	\$745,939	\$324,765
<i>QALY decomposition</i>			
RF state	10.17	4.60	5.57
LR state	1.99	1.32	0.67
DM state	2.00	1.73	0.27
Total QALYs	14.16	7.65	6.52

3.3 External and Structural Validation

Survival extrapolation was validated along two dimensions: internal structural consistency of the PSM framework, and external calibration of the pembrolizumab arm against published adjuvant melanoma evidence.

Structural consistency. Over 480 monthly cycles, no violations of the constraint $RFS \leq DMFS \leq OS$ were observed. All health-state probabilities remained non-negative, and the sum of state probabilities was within 10^{-12} of unity at every cycle. The general population mortality hazard floor, applied from 60 months onward, ensures that the model’s long-term survival does not exceed age- and sex-matched US life-table survival at any time point. At

30 years, the model projects combo OS of 21.5% and pembro OS of 10.9%, both below the general-population survival of 22.1% at 30 years for the modeled cohort (baseline age 61, pooled 64% male), consistent with the hazard-floor property that a cohort with a history of excess mortality remains below the general-population curve.

External validation. The pembrolizumab-alone arm predictions were compared against published long-term landmark values from two external adjuvant melanoma trials [25, 26]. At 5 years, the model predicted RFS of 40.5%, DMFS of 53.8%, and OS of 74.5% for the pembrolizumab arm. The predicted 5-year OS (74.5%) was consistent with the observed KEYNOTE-942 value (71.3%) and with the 5-year OS of 76% in CheckMate 238 [27]. At 7 years, the model predicted RFS of 31.9% and DMFS of 44.9%, below the KEYNOTE-054 7-year values (RFS 50%, DMFS 54%) [25]; the absolute differences were -18.1 and -9.1 percentage points, respectively. The 5-year RFS of 40.5% was -10.5 percentage points relative to the CheckMate 238 5-year RFS of 51%. This is consistent with the higher-risk population enrolled in KEYNOTE-942 (stage IIIB–IV including resected stage IV vs IIIA–C in KEYNOTE-054). Only published landmark values were used; no external trial extrapolations were employed. Because the pembrolizumab arm serves as the comparator for the incremental analysis, its under-prediction of RFS may, if anything, overstate the incremental QALY gain attributable to intismeran; however, the direction of any bias is not straightforward, as the model’s predicted OS for the pembrolizumab arm was close to observed (74.5% vs 71.3% at 5 years). No external long-term data exist for the combination arm (first-in-class therapy). Instead of relying on this comparison to calibrate the base case, scenario analyses that directly bound the treatment effect — treatment-effect waning and no-direct-OS-benefit (Section 3.5) — were used to bound the range of plausible ICER estimates.

3.4 Probabilistic and Deterministic Uncertainty

Probabilistic sensitivity analysis. Probabilistic sensitivity analysis across 6,000 structurally valid iterations (0 rejections) quantified joint parameter uncertainty. The probability of cost-effectiveness increased with the willingness-to-pay threshold, from 52.4% at \$50,000/QALY to

93.4% at \$100,000/QALY and 98.4% at \$150,000/QALY. Figure 2 presents the cost-effectiveness acceptability curve across the full range of willingness-to-pay thresholds. All 6,000 iterations produced positive incremental QALYs, and 11.2% were cost-saving. Because cost-saving draws have an undefined ICER, incremental net monetary benefit and probability of cost-effectiveness are the preferred summary statistics (Table 3) [23].

Table 3: Probabilistic Sensitivity Analysis

Metric	Mean	Median	95% CI
Incremental cost	\$293,238	\$321,291	– \$533,282 to \$825,708
Incremental QALYs	6.65	6.51	4.13–9.83
NMB @ \$100,000/QALY	\$371,570	—	—
NMB @ \$150,000/QALY	\$703,975	—	—

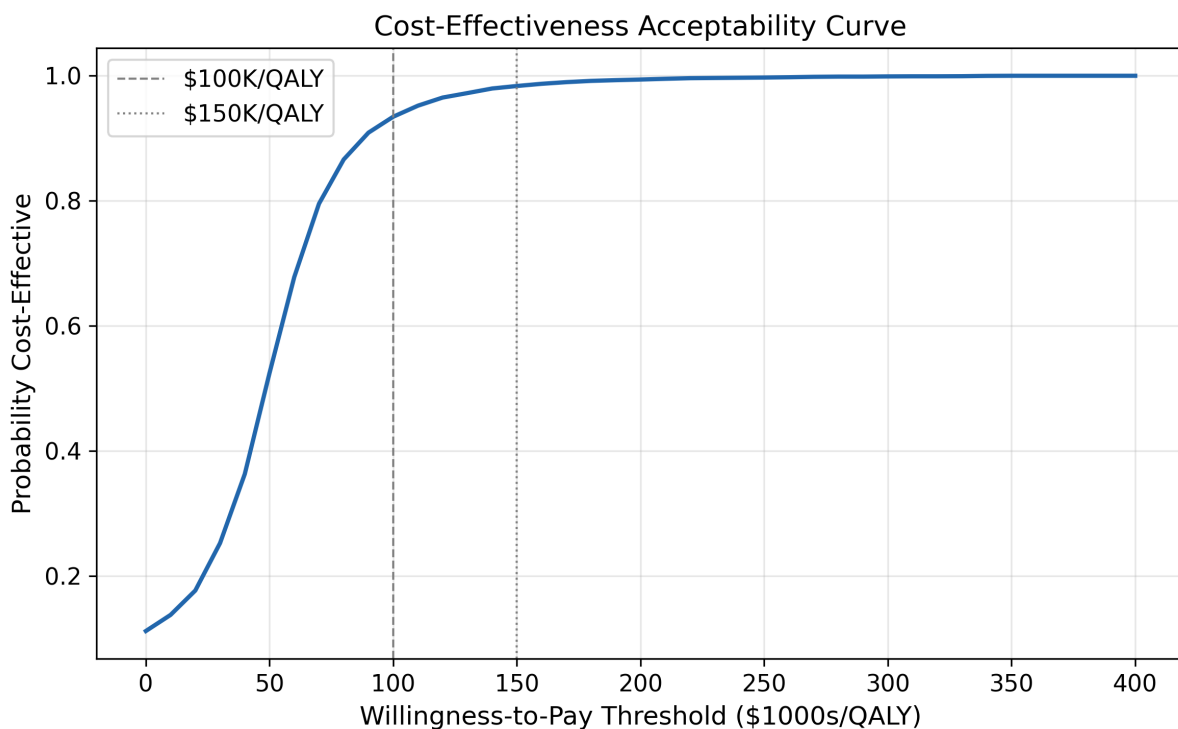


Figure 2: Cost-effectiveness acceptability curve.

Deterministic sensitivity analysis. One-way sensitivity analysis identified the parameters with the greatest influence on the base-case ICER (Supplementary Table S11). As shown in Figure 3, the ICER was most sensitive to the intismeran acquisition price (range \$43,702–

\$55,979/QALY across \$160,000–\$240,000) and the discount rate (\$49,330–\$53,363/QALY across 0%–5%), with a smaller contribution from the pembrolizumab OS parameter and downstream management costs; the ICER remained below \$61,000/QALY across all tested parameter ranges.

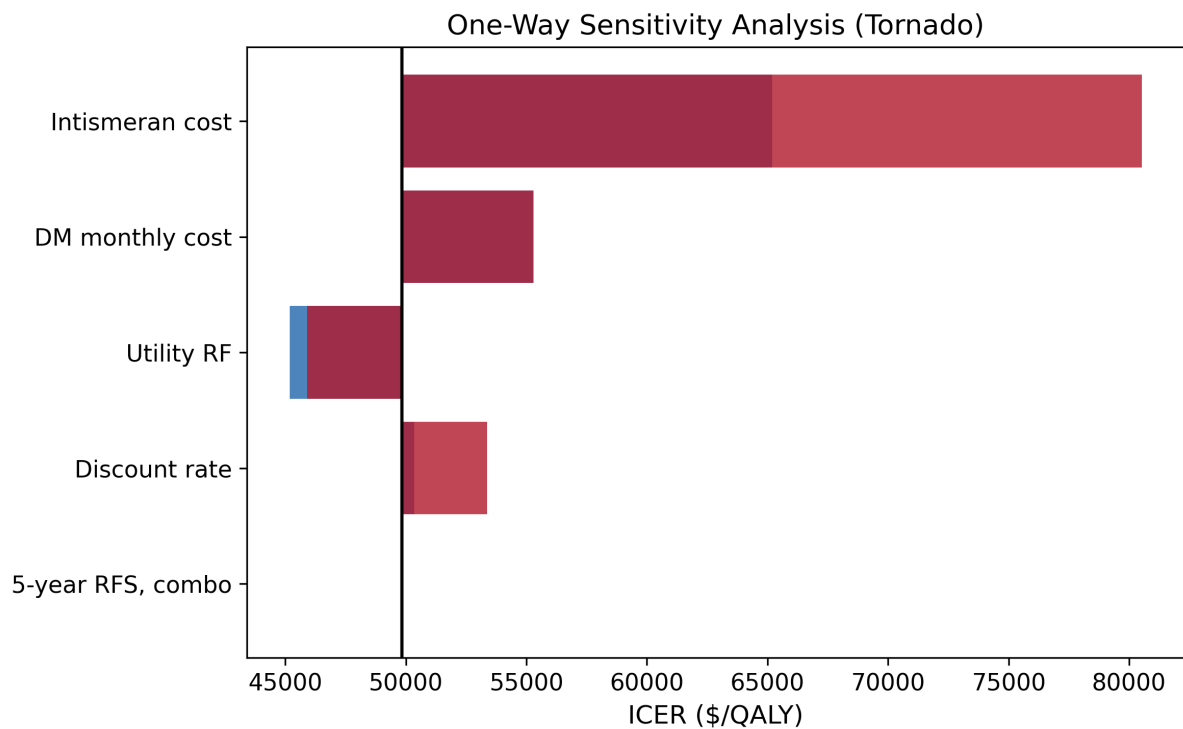


Figure 3: Tornado diagram of one-way sensitivity analysis.

3.5 Structural Scenario Analyses

Scenario analysis results are summarized in Table 4.

Table 4: Scenario and Deterministic Threshold Acquisition Price Analyses

Scenario	Δ Cost	Δ QALY	ICER	NMB @ \$150K
Base case (GP-constrained)	\$324,765	6.52	\$49,841	\$652,643
<i>Pricing</i>				
Price: \$100K/course	\$224,765	6.52	\$34,494	\$752,643
Price: \$300K/course	\$424,765	6.52	\$65,187	\$552,643
Price: \$500K/course	\$624,765	6.52	\$95,881	\$352,643
<i>Alternative assumptions</i>				
Weibull OS	\$503,437	8.40	\$59,959	\$756,008
Discount rate: 0%	\$604,444	12.25	\$49,330	\$1,233,503
Discount rate: 5%	\$240,037	4.50	\$53,363	\$434,691
Time horizon: 10 years	\$97,868	1.34	\$73,233	\$102,591
Time horizon: 20 years	\$147,215	3.52	\$41,785	\$381,256
General population utilities	\$324,765	6.25	\$51,979	\$612,435
<i>Treatment-effect persistence</i>				
Recurrence convergence (5–8y)	\$39,862	2.09	\$19,055	\$273,926
Hazard convergence (5–10y)	\$249,355	2.39	\$104,247	\$109,439
No direct OS benefit	–\$22,543	0.52	Dominant	\$100,121

3.6 Deterministic Threshold Acquisition Price

The deterministic threshold acquisition price of intismeran was estimated by identifying the acquisition price at which the ICER reached conventional willingness-to-pay thresholds. At the assumed base-case acquisition price of \$200,000 per course, the ICER was \$49,841/QALY. The deterministic threshold acquisition price was approximately \$527,000 per course at a willingness-to-pay threshold of \$100,000/QALY and approximately \$852,000 per course at \$150,000/QALY. Figure 4 illustrates the resulting price–ICER relationship and the intersection of the curve with these two willingness-to-pay thresholds.

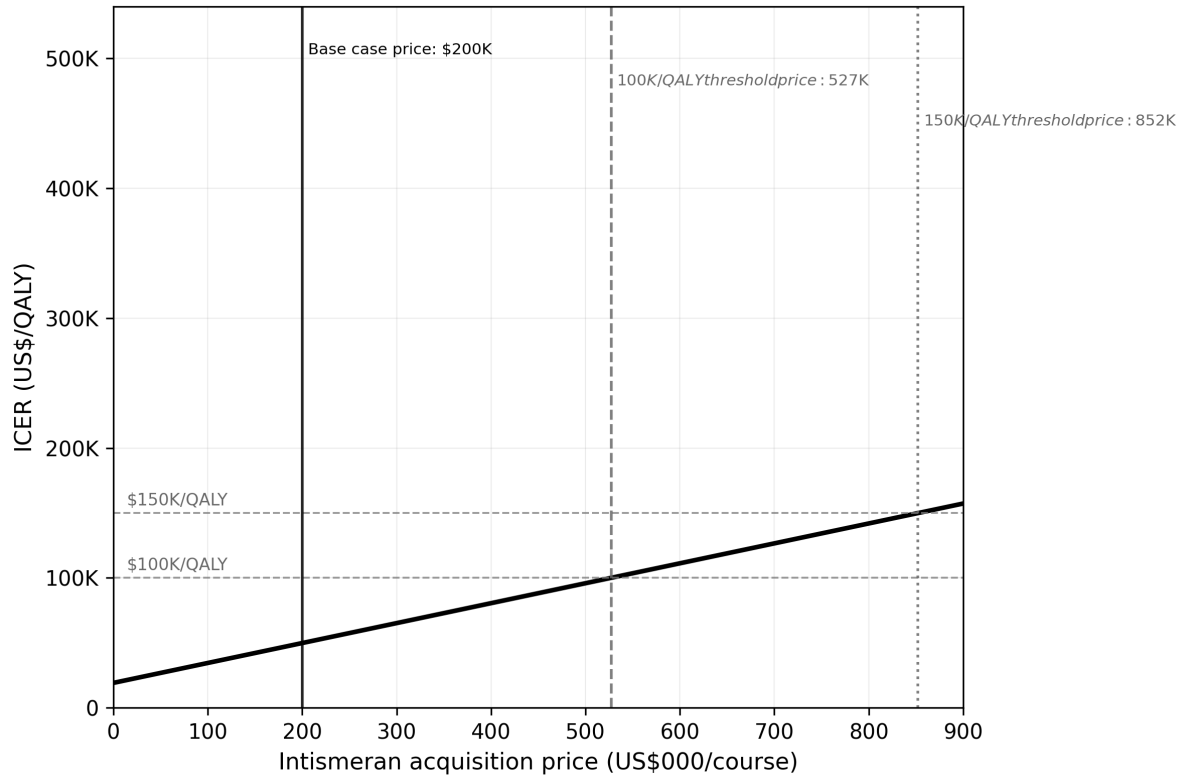


Figure 4: Price–ICER curve and deterministic threshold pricing. Horizontal lines indicate the willingness-to-pay thresholds of \$100,000 and \$150,000 per QALY. Vertical reference lines indicate the base-case price (\$200,000) and the deterministic threshold prices at \$100,000/QALY and \$150,000/QALY.

4 Discussion

4.1 Principal Findings

This early economic evaluation demonstrates that intismeran plus pembrolizumab was projected to be cost-effective under the base-case assumptions for adjuvant treatment of resected stage IIIB–IV melanoma at conventional US willingness-to-pay thresholds. The base case ICER of \$49,841/QALY, the 6.52 QALY gain, and the 93.4% probability of cost-effectiveness at \$100,000/QALY (98.4% at \$150,000/QALY) all support this conclusion. Notably, under the modeled recurrence effects and health-state cost and utility assumptions, the combination remained dominant (lower cost and higher QALY) in the scenario where intismeran provided no direct OS benefit beyond its effect on recurrence. This scenario indicates that favorable economics can be supported by recurrence prevention alone within the model, rather than es-

tablishing that the conclusion is independent of OS assumptions.

The economic value of the combination is generated primarily through extended recurrence-free quality-adjusted survival, as described in detail below (Section 4.2). The ICER is notably lower than the \$75,206 from the prior exploratory analysis [8] (Section 4.5), reflecting the combined effect of mature clinical data and a structurally constrained modelling framework.

4.2 Economic Drivers and Cost Consequences

The incremental \$324,765 lifetime cost of intismeran plus pembrolizumab exceeds the \$200,000 acquisition cost of the intismeran course, because combination-arm patients survive substantially longer and thus accumulate more time in the locoregional-recurrence and distant-metastasis health states (combo LR management \$112,203 vs pembro \$74,477; combo DM management \$524,121 vs pembro \$453,082). The QALY decomposition (Table 2) shows that the 6.52 incremental QALYs are driven almost entirely by additional time in the recurrence-free health state: the combination arm generates 5.57 more RF QALYs, with a modest additional 0.67 LR QALYs and 0.27 DM QALYs reflecting longer overall survival. This asymmetry — a large QALY gain concentrated in the highest-utility health state, outweighing the additional costs from extended survival — is the structural mechanism behind the favourable ICER.

The economic value is driven by a single mechanism: extended quality-adjusted survival. The combination arm's longer survival produces a 6.52-QALY advantage concentrated in the recurrence-free state (5.57 RF QALYs gained). This occurs **despite** higher downstream management costs (\$108,765 net: \$37,726 LR + \$71,039 DM), which reflect longer survival rather than a cost-offset mechanism. The favourable ICER is therefore driven entirely by the magnitude of the QALY gain, not by avoidance of downstream disease-management costs.

4.3 Pricing and Reimbursement Implications

The base case assumes an intismeran acquisition price of \$200,000 per course, based on Jefferies analyst estimates. At this price, the ICER of \$49,841/QALY is well below conventional US thresholds. The deterministic threshold price analysis (Section 3.6) shows that the ICER remains below \$100,000/QALY at intismeran prices up to approximately \$527,000 per course

and below \$150,000/QALY up to approximately \$852,000 per course, providing a deterministic cost-effectiveness benchmark under the base-case assumptions.

While individualized production requirements — including tumor sequencing, mRNA synthesis, lipid nanoparticle formulation, and quality assurance for each batch — may contribute to acquisition-price pressure, production cost and commercial price are distinct quantities. The \$200,000 estimate reflects the expected market price at launch, not the manufacturer's cost of goods sold. Several factors may reduce future production costs: sequencing costs continue to decline [28], mRNA platform manufacturing is becoming increasingly standardized, and automation of the synthesis and formulation pipeline could reduce per-batch costs. However, whether these technical efficiencies will translate into lower acquisition prices for payers is uncertain, as pricing decisions are influenced by clinical value, competitive dynamics, and reimbursement negotiations, not solely by production economics.

Cost-effectiveness should not be conflated with affordability. The deterministic threshold price of approximately \$527,000 per course at \$100,000/QALY implies that the combination could be considered cost-effective at more than twice the current estimated price, but the incremental per-patient budget impact of adding \$200,000 in drug costs to an already expensive pembrolizumab regimen would be substantial in a population of eligible patients. Formal budget-impact analysis is needed alongside cost-effectiveness evidence for reimbursement decision-making.

4.4 Long-Term Survival and Structural Uncertainty

Long-term survival extrapolation is the dominant source of structural uncertainty in this analysis. The mature 5-year RFS and DMFS data from KEYNOTE-942 provide a robust foundation for the recurrence and metastasis components of the model, but OS remains exploratory (HR 0.471; 95% CI 0.165–1.345; 7 events per arm) and must be extrapolated to a lifetime horizon.

The necessity of the general population mortality hazard floor is best illustrated by comparing the constrained and unconstrained models. Without the GP mortality floor, the log-normal survival fit for the combination arm predicts a 30-year OS of 79.1%, far exceeding the age- and sex-matched general population survival of 22.1% at 30 years for the modeled cohort.

This is implausible: a cohort of patients with resected stage IIIB–IV melanoma cannot have a lower mortality rate than the general population. Applying the GP mortality hazard floor from 60 months onward — a standard practice recommended by NICE DSU TSD 14 [13] and used in published melanoma CEAs [10] — reduces the 30-year combination OS to 21.5%, a clinically plausible trajectory. The hazard floor reduces the ICER from \$51,753 (unconstrained) to \$49,841 (GP-constrained); the direction of this change is non-intuitive (the constraint reduces the ICER) because the unconstrained model predicts implausibly high survival for both arms, with a larger absolute gain for the combination arm, and the hazard floor compresses that difference while preserving the within-trial benefit.

The OS uncertainty is further underscored by the treatment-effect waning scenarios. In the recurrence-convergence scenario (5→8y), the ICER falls to \$19,055/QALY because the recurrence-related QALY advantage compresses from 6.52 to 2.09, while the \$200,000 imtizeran acquisition cost is preserved – the incremental cost therefore drops from \$324,765 to \$39,862 faster than QALYs, producing a lower ICER. The hazard-convergence scenario (5→10y) yields the highest ICER of \$104,247/QALY. These two scenarios bracket the range of plausible ICER estimates under alternative treatment-persistence assumptions.

Finally, the choice of parametric distribution, while important, is secondary to the structural scenarios. The log-normal distribution was selected by AIC for the combination OS curve and was applied uniformly across all components for parsimony. However, AIC measures fit within the observed 5-year follow-up; it does not establish long-term clinical validity. The Weibull alternative, which produced an ICER of \$59,959/QALY, also satisfied the structural constraints and is a plausible alternative. The most informative bounds on the ICER come not from alternative parametric forms but from the treatment-effect waning and no-direct-OS-benefit scenarios, which directly address the question: what if the treatment effect does not persist? The best AIC-ranked mixed-family combination (OS: log-normal/Weibull, r_{DM} : log-normal/log-logistic, r_{LR} : log-logistic/Weibull) produced an ICER of \$61,102, consistent with the Weibull scenario and within the range of structural uncertainty, further supporting the robustness of the base-case conclusion.

4.5 Comparison with Previous Economic Evaluations

Direct comparison of ICERs across adjuvant melanoma cost-effectiveness studies is not straightforward, because the incremental decision differs. Bensimon et al. [11] and Zhang et al. [10] evaluated pembrolizumab versus observation, estimating the value of adding immunotherapy to surveillance. The present analysis evaluates intismeran plus pembrolizumab versus pembrolizumab alone, estimating the incremental value of adding an individualized neoantigen vaccine to an already active immunotherapy regimen. The relevant policy question is therefore not whether the combination is cost-effective in absolute terms, but whether the combination of two active agents can provide sufficient additional clinical value to justify the acquisition cost of the second agent and the additional lifetime disease-management costs associated with prolonged survival.

The ICER of \$49,841/QALY is within the range of published estimates for the initial pembrolizumab decision (\$38,050–\$78,400/QALY), but this comparability is coincidental rather than meaningful. The absolute QALY gain from adding intismeran to pembrolizumab (6.52 QALYs) is larger than the gain from adding pembrolizumab to observation (1.2–2.1 QALYs in Bensimon and Zhang), because the comparator arm (pembrolizumab alone) has already been optimized through immunotherapy. The incremental cost is also larger (\$324,765 vs approximately \$100,000–\$150,000). The ratio happens to fall within a similar range because both numerator and denominator increase proportionally, not because the value proposition is comparable.

The comparison with the 2024 exploratory analysis [8] is more informative but requires careful interpretation. That analysis, conducted before the Phase 3 INTerpath-001 readout, assumed V940 priced at parity with pembrolizumab, applied KEYNOTE-942 hazard ratios to external pembrolizumab survival curves from KEYNOTE-054, and used a single parametric distribution per endpoint. The present analysis differs on all four dimensions: post-Phase III evidence context, direct fitting to KEYNOTE-942 5-year data, a \$200,000 price assumption, and a ratio-constrained structural framework with a general population mortality floor. The lower ICER (\$49,841 vs \$75,206) is the net result of these four differences, not a univariate effect of any single factor, and should not be interpreted as evidence that intismeran is simply

more effective than previously estimated.

4.6 Clinical and External Validity

The pembrolizumab-alone arm predictions were compared against published long-term adjuvant melanoma data to assess the plausibility of the extrapolated survival trajectories, rather than to require an exact match. The KEYNOTE-942 trial enrolled a higher-risk population (stage IIIB–IV, including resected stage IV) than KEYNOTE-054 (stage IIIA–C) and Check-Mate 238 (stage IIIB–IV, but with different entry criteria). The model’s lower 7-year RFS for the pembrolizumab arm (31.7% vs 50% in KEYNOTE-054) is therefore expected and reflects the underlying risk differences rather than a failure of calibration. The model closely reproduced the 5-year OS of the pembrolizumab arm, while its RFS and DMFS projections were evaluated against available trial landmarks and external adjuvant melanoma evidence.

No external long-term data exist for the combination arm (first-in-class therapy). The positive Phase 3 INTERpath-001 readout provides directional confirmation of the clinical benefit of intismeran plus pembrolizumab, but the numerical hazard ratios have not yet been disclosed, and the extent to which Phase 3 results will align with the Phase 2b KEYNOTE-942 estimates used in this model remains to be determined. Two structural scenarios address the uncertainty in the combination arm: (i) the treatment-effect waning scenarios — recurrence-convergence at 5→8y (ICER \$19,055/QALY, most favourable) and hazard-convergence at 5→10y (ICER \$104,247/QALY, least favourable); and (ii) the no-direct-OS-benefit scenario, an OS-neutral structural scenario in which intismeran plus pembrolizumab remained dominant (lower cost and higher QALY) compared with pembrolizumab alone, under the modeled recurrence effects and health-state cost and utility assumptions, indicating that recurrence prevention alone can support favorable economics within the model but does not establish independence from the model assumptions.

4.7 Limitations

This analysis has several limitations that should be considered when interpreting the results. First, this early economic evaluation derives efficacy inputs from Phase 2b KEYNOTE-942

(n=157). Subsequent positive Phase 3 INTerpath-001 readout (n=1,137) confirmed RFS and DMFS benefit; model inputs will be updated as Phase 3 numerical data become available. Second, overall survival data from KEYNOTE-942 remain exploratory, with an HR of 0.471 (95% CI 0.165–1.345) and only 7 events per arm at the 5-year analysis, and the 5-year OS estimate of 92.2% versus 71.3% should be interpreted with caution. Third, survival curves were reconstructed from published Kaplan–Meier figures using digitized coordinates rather than being estimated from individual patient-level data, which introduces measurement error in the parametric fits. Fourth, survival extrapolation beyond the observed 5-year follow-up, while structurally constrained and anchored to a general population mortality floor, remains sensitive to the choice of parametric distribution and the assumption that the treatment effect persists indefinitely. Time horizon sensitivity analysis showed that the ICER remained below conventional US willingness-to-pay thresholds at 20, 30, and 40 years (\$41,785, \$45,349, and \$49,841/QALY, respectively), although the point estimate increased with longer extrapolation, indicating that the 40-year extrapolation window did not materially drive the base-case results. Fifth, the intismeran acquisition price of \$200,000 per course is based on analyst estimates, and the actual market price may differ. Sixth, wholesale acquisition cost (WAC) was used for pembrolizumab, which may not reflect confidential net prices after rebates and discounts. Because pembrolizumab costs were applied symmetrically to both arms, plausible reductions in its net price would have limited impact on the incremental cost comparison. Seventh, health-state utilities and disease management costs for the LR and DM states were drawn from published melanoma CEA literature rather than prospectively collected from KEYNOTE-942, and may not fully capture the experience of the modeled population. Eighth, indirect costs, productivity losses, and informal caregiver burden were not included, which may underestimate the societal value of preventing recurrence and distant metastasis in a working-age population. Ninth, the KEYNOTE-942 trial enrolled a mixed-stage population (IIIB–IV), and baseline recurrence risk varies substantially by disease stage. The Phase 2b sample size precluded formal subgroup cost-effectiveness analysis by stage, PD-L1 status, or other risk factors. If the treatment effect differs by baseline risk, the average ICER may not reflect the value of intismeran in specific subgroups, and reimbursement decisions would ideally consider stage-specific estimates. The

Phase 3 INTERpath-001 trial (N=1,137, stage IIB–IV) will provide sufficient statistical power for subgroup analyses, and the present findings should be updated when those data become available. Tenth, no formal budget-impact analysis was conducted; the cost-effectiveness findings should be complemented by an affordability assessment when pricing and population-level data become available. Finally, this analysis is specific to the resected stage IIB–IV melanoma population of KEYNOTE-942 and may not be generalizable to other melanoma stages or tumor types, although the Phase 3 INTERpath-001 trial enrolled a broader stage IIB–IV population.

4.8 Conclusion

Intismeran plus pembrolizumab provides clinical benefit in adjuvant melanoma and was projected to be cost-effective at the estimated acquisition cost of \$200,000 per course under the base-case assumptions. The deterministic threshold price analysis shows that the ICER remains below \$100,000/QALY at intismeran prices up to approximately \$527,000 per course, providing a deterministic cost-effectiveness benchmark under the base-case assumptions. The large QALY gain from extended recurrence-free survival represents clinical benefit, while the incremental cost reflects the acquisition price plus additional disease-management expenditure from longer survival. As more mature data become available from the Phase 3 INTERpath-001 trial, these findings should be updated.

Declarations

Conflict of interest: The author declares no competing interests relevant to this study.

Reporting: This study adheres to the Consolidated Health Economic Evaluation Reporting Standards 2022 (CHEERS 2022) [1] (Supplementary Checklist).

Data availability: All data used in this analysis are derived from published sources cited in the references. The model code is available at <https://github.com/yichao2022/cea-intismeran>.

Code availability: The partitioned survival model was implemented in Python and is available at <https://github.com/yichao2022/cea-intismeran>.

Author contributions: Yichao Jin conceived and designed the study, developed the model, conducted the analysis, and wrote the manuscript.

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Supplementary Appendix

Cost-Effectiveness of Intismeran Autogene (mRNA-4157/V940) plus Pembrolizumab versus Pembrolizumab Alone as Adjuvant Therapy for Resected Stage IIIB–IV Melanoma

Yichao Jin, PhD

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1 Model Structure and Assumptions

1.1 Health-State Definitions

Table S1: Health-State Definitions, Clinical Interpretation, and Model Inputs

State	Clinical Definition	Utility	Monthly Cost
RF	Recurrence-free after complete resection; receiving adjuvant pembrolizumab (up to 18 cycles) with or without ipilimumab (up to 9 doses)	0.83	\$0
LR	Confirmed locoregional recurrence without distant metastasis; includes salvage surgery, re-treatment, and surveillance	0.64	\$3,000
DM	Confirmed stage IV melanoma with distant metastasis; includes systemic therapy, palliative care, and monitoring	0.55	\$12,000
Death	All-cause mortality	0	\$0

Note: Utilities and costs are drawn from published melanoma CEA literature [1–3]. Costs are in 2026 USD. See Supplementary Table S9 for full PSA distribution parameters.

1.2 State Occupancy Equations

State occupancy at each monthly cycle is determined by the partitioned survival model (PSM) approach, using three parametric survival functions:

- **RFS(t)** — Recurrence-free survival: probability of remaining alive and free of any recurrence (locoregional or distant) at time t .
- **DMFS(t)** — Distant metastasis-free survival: probability of remaining alive and free of distant metastasis at time t .
- **OS(t)** — Overall survival: probability of being alive at time t .

The occupancy equations are derived from the ordering of these three events:

- All patients who are recurrence-free are also distant metastasis-free and alive: $RF = RFS$.
- Patients who have had a locoregional recurrence but have not yet developed distant metastasis are those who are distant metastasis-free minus those who are recurrence-free: $LR = DMFS - RFS$.
- Patients who are alive with distant metastasis are those who are alive minus those who are distant metastasis-free: $DM = OS - DMFS$.
- The dead state is the complement of overall survival: $Dead = 1 - OS$.

The constraint $RFS \leq DMFS \leq OS$ is enforced at all time points to ensure non-negative state occupancy. This ordering is clinically meaningful: a patient cannot have distant metastasis without having had a recurrence event ($RFS \leq DMFS$), and a patient cannot die without having both events ($DMFS \leq OS$). The ratio-constrained structure $r_{LR} = RFS/DMFS$ and $r_{DM} = DMFS/OS$ guarantees this ordering by construction.

1.3 Model Schematic

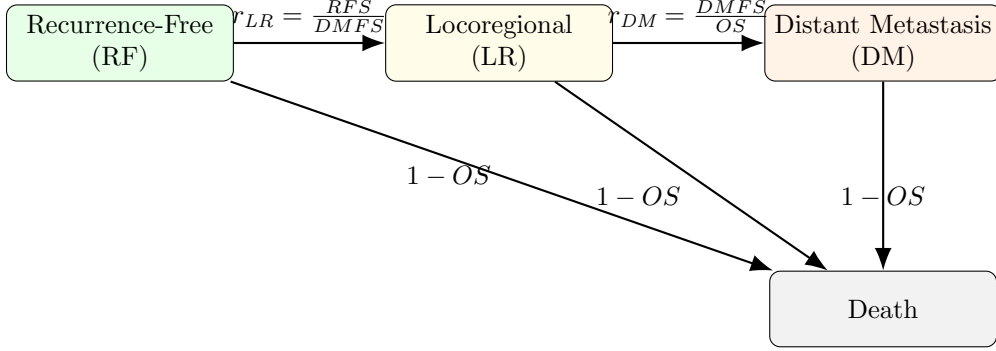


Figure S1: Partitioned survival model structure. The model uses four mutually exclusive health states. State occupancy is derived from three parametric survival functions (RFS, DMFS, OS) rather than from transition probabilities. Arrows denote the ratio-constrained relationship $RFS \leq DMFS \leq OS$.

2 Survival Data and Parametric Extrapolation

2.1 Extracted Survival Probabilities

Survival probabilities were extracted from the published Kaplan–Meier curves (Figure 1) of Khattak et al. (2026) [4] using WebPlotDigitizer v4.7. Data were extracted at 18, 24, 36, 48, and 60 months for OS and RFS, and at 18, 24, 36, and 48 months for DMFS. The extracted values are reported in Table S2.

Table S2: Extracted KEYNOTE-942 Survival Probabilities (%)

Month	RFS (%)		DMFS (%)		OS (%)	
	Combo	Pembro	Combo	Pembro	Combo	Pembro
18	80.4	62.2	92.0	73.0	96.0	93.4
24	78.3	60.0	90.8	70.6	96.0	93.4
36	73.8	55.6	85.6	65.4	93.8	85.6
48	72.4	49.1	83.9	65.4	92.2	85.6
60	68.8	49.1	—	—	92.2	71.3

Digitization method: WebPlotDigitizer v4.7, manual extraction. The 60-month landmark values match the published 5-year ASCO 2026 update [4]. At earlier time points, the 18-month RFS rates from the primary analysis [5] differ from digitized values by 1.4 pp (combo) and 0.2 pp (pembro), consistent with typical digitization error. The 5-year RFS 95% CIs: combo 56.3–78.3%, pembro 33.3–63.0%.

3 Clinical Input Provenance

Table S3 summarizes the provenance of all clinical inputs used in the model. The evidence status column classifies each input into one of five categories:

- **Published:** The value is directly reported in the cited source.
- **Digitized:** The value was extracted from a published Kaplan–Meier curve using Web-PlotDigitizer v4.7.
- **Derived:** The value is mathematically derived from other inputs (e.g., ratio of survival functions).
- **Modeled:** The value is produced by the model’s parametric extrapolation (e.g., long-term survival beyond observed data).
- **Topline only:** The source reports a statistically significant result but the numerical hazard ratio or landmark survival probabilities have not yet been disclosed.

Table S3: Clinical Input Provenance

Input	Value	Evidence Status	Source
Input	Value	Evidence Status	Source
<i>KEYNOTE-942 (Phase 2b, 5-year follow-up)</i>			
Recurrence-Free Survival (RFS)			
18-month RFS, combo	80.4%	Digitized	Fig. 1, Khattak 2026
24-month RFS, combo	78.3%	Digitized	Fig. 1, Khattak 2026
36-month RFS, combo	73.8%	Digitized	Fig. 1, Khattak 2026
48-month RFS, combo	72.4%	Digitized	Fig. 1, Khattak 2026
60-month RFS, combo	68.8%	Published	Khattak 2026
60-month RFS, combo 95% CI	56.3–78.3%	Published	Khattak 2026
18-month RFS, pembro	62.2%	Digitized	Fig. 1, Khattak 2026
24-month RFS, pembro	60.0%	Digitized	Fig. 1, Khattak 2026
36-month RFS, pembro	55.6%	Digitized	Fig. 1, Khattak 2026
48-month RFS, pembro	49.1%	Digitized	Fig. 1, Khattak 2026
60-month RFS, pembro	49.1%	Published	Khattak 2026
60-month RFS, pembro 95% CI	33.3–63.0%	Published	Khattak 2026
Distant Metastasis-Free Survival (DMFS)			
18-month DMFS, combo	92.0%	Digitized	Fig. 1, Khattak 2026
24-month DMFS, combo	90.8%	Digitized	Fig. 1, Khattak 2026
36-month DMFS, combo	85.6%	Digitized	Fig. 1, Khattak 2026
48-month DMFS, combo	83.9%	Digitized	Fig. 1, Khattak 2026
60-month DMFS, combo	–	Not reported	N/A
18-month DMFS, pembro	73.0%	Digitized	Fig. 1, Khattak 2026
24-month DMFS, pembro	70.6%	Digitized	Fig. 1, Khattak 2026
36-month DMFS, pembro	65.4%	Digitized	Fig. 1, Khattak 2026
48-month DMFS, pembro	65.4%	Digitized	Fig. 1, Khattak 2026
60-month DMFS, pembro	–	Not reported	N/A
Overall Survival (OS)			

Input	Value	Evidence Status	Source
18-month OS, combo	96.0%	Digitized	Fig. 1, Khattak 2026
24-month OS, combo	96.0%	Digitized	Fig. 1, Khattak 2026
36-month OS, combo	93.8%	Digitized	Fig. 1, Khattak 2026
48-month OS, combo	92.2%	Published	Khattak 2026
60-month OS, combo	92.2%	Published	Khattak 2026
18-month OS, pembro	93.4%	Digitized	Fig. 1, Khattak 2026
24-month OS, pembro	93.4%	Digitized	Fig. 1, Khattak 2026
36-month OS, pembro	85.6%	Digitized	Fig. 1, Khattak 2026
48-month OS, pembro	85.6%	Published	Khattak 2026
60-month OS, pembro	71.3%	Published	Khattak 2026
Hazard Ratios			
RFS HR (combo vs. pembro)	0.510	Published	Khattak 2026
RFS HR 95% CI	0.294–0.887	Published	Khattak 2026
DMFS HR (combo vs. pembro)	0.411	Published	Khattak 2026
DMFS HR 95% CI	0.200–0.843	Published	Khattak 2026
OS HR (combo vs. pembro)	0.471	Published	Khattak 2026
OS HR 95% CI	0.165–1.345	Published	Khattak 2026
OS events	14 (7 per arm)	Published	Khattak 2026
<i>INTerpath-001 (Phase 3, topline announcement)</i>			
RFS	Statistically significant	Topline only	Merck/Moderna 2026
DMFS	Statistically significant	Topline only	Merck/Moderna 2026
Numerical HR / KM curves	Not yet disclosed	Topline only	N/A
<i>Modeled / Derived Values</i>			
Fitted log-normal parameters			
OS μ (combo), σ (combo)	4.825, 0.879	Derived	Least-squares fit
OS μ (pembro), σ (pembro)	4.214, 1.332	Derived	Least-squares fit
r_DM μ (combo), σ (combo)	4.285, 0.851	Derived	Ratio fit
r_DM μ (pembro), σ (pembro)	3.475, 1.390	Derived	Ratio fit
r_LR μ (combo), σ (combo)	3.420, 1.250	Derived	Ratio fit
r_LR μ (pembro), σ (pembro)	2.268, 1.749	Derived	Ratio fit
Long-term extrapolation			
10-year OS, combo	74.4%	Modeled	Log-normal extrapolation
10-year OS, pembro	46.3%	Modeled	Log-normal extrapolation
20-year OS, combo (GP-constrained)	18.3%	Modeled	GP hazard floor
20-year OS, pembro (GP-constrained)	7.8%	Modeled	GP hazard floor
40-year OS, combo (GP-constrained)	0.0%	Modeled	GP hazard floor

3.1 Observed vs. Fitted Survival Curves

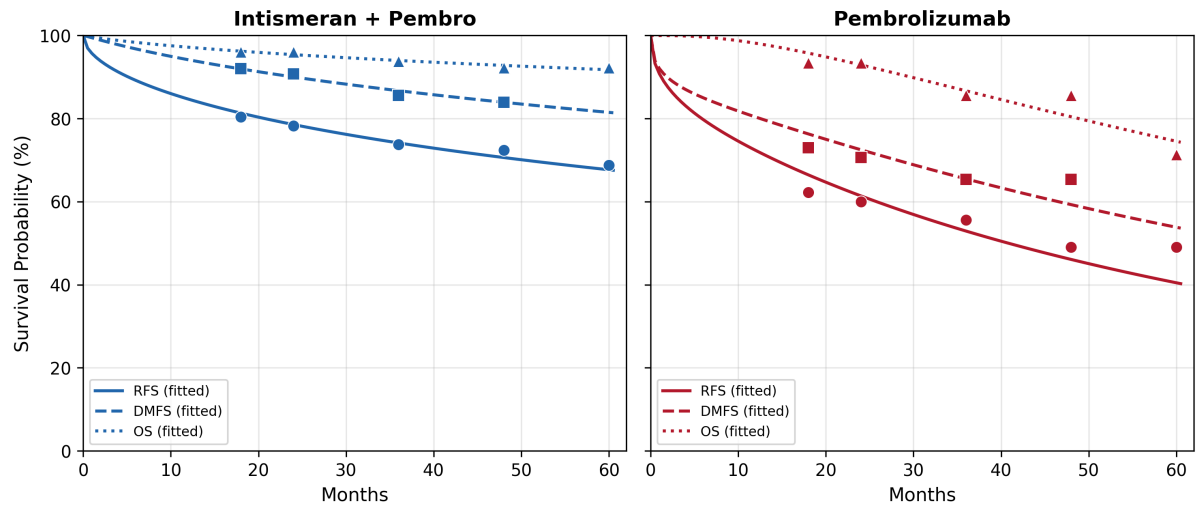


Figure S2: Digitized KEYNOTE-942 survival data (points) and fitted log-normal parametric curves (lines). Left panel: intismeran plus pembrolizumab arm. Right panel: pembrolizumab alone arm.

4 Survival Model Selection

4.1 Component-Level Parametric Fit

Table S4: Component-Level Parametric Fit Across Five Distributions

Component	Distribution	AIC	RMSE	Rank
<i>OS – Combo arm</i>				
Exponential	1 param	-45.49	0.0087	4
Weibull	2 param	-49.84	0.0046	2
Log-normal	2 param	-50.08	0.0045	1
Log-logistic	2 param	-49.89	0.0046	3
Generalized gamma	3 param	-47.97	0.0045	5
<i>OS – Pembro arm</i>				
Exponential	1 param	-31.27	0.0359	4
Weibull	2 param	-32.00	0.0273	1
Log-normal	2 param	-31.16	0.0297	3
Log-logistic	2 param	-31.69	0.0282	2
Generalized gamma	3 param	-30.33	0.0264	5
<i>r_{DM} – Combo arm</i>				
Exponential	1 param	-36.79	0.0078	3
Weibull	2 param	-36.53	0.0063	4
Log-normal	2 param	-36.93	0.0060	1
Log-logistic	2 param	-36.61	0.0062	2
Generalized gamma	3 param	-34.74	0.0061	5
<i>r_{DM} – Pembro arm</i>				
Exponential	1 param	-19.41	0.0688	5
Weibull	2 param	-28.43	0.0174	3
Log-normal	2 param	-27.23	0.0202	4
Log-logistic	2 param	-30.30	0.0137	2
Generalized gamma	3 param	-32.45	0.0082	1
<i>r_{LR} – Combo arm</i>				
Exponential	1 param	-30.38	0.0392	5
Weibull	2 param	-38.97	0.0136	3
Log-normal	2 param	-39.49	0.0129	2
Log-logistic	2 param	-39.70	0.0126	1
Generalized gamma	3 param	-38.43	0.0118	4
<i>r_{LR} – Pembro arm</i>				
Exponential	1 param	-31.82	0.0340	5
Weibull	2 param	-33.60	0.0233	1
Log-normal	2 param	-33.37	0.0238	2
Log-logistic	2 param	-33.51	0.0235	3
Generalized gamma	3 param	-31.69	0.0231	4

4.2 Joint Model Selection

A total of 4,096 distribution combinations were evaluated for structural validity at 40 years. All 4,096 satisfied the $RFS \leq DMFS \leq OS$ ordering constraint (0 crossing violations). The full results are available online ([output/survival_combos_4096.csv](#)).

Table S5: Top 10 Joint Model Combinations by Total AIC

Rank	OS	r_{DM}	r_{LR}	Total AIC	OS40c	OS40p
	C/P	C/P	C/P			
1	LN / W	LN / LL	LL / W	-222.61	76.3%	0.0%
2	LN / W	LN / LL	LL / LL	-222.52	76.3%	0.0%
3	LN / W	Exp / LL	LL / W	-222.47	76.3%	0.0%
4	LLog / W	LN / LL	LL / W	-222.42	72.9%	0.0%
5	LN / W	LN / LL	LN / W	-222.40	76.3%	0.0%
6	LN / W	LN / LL	LL / LN	-222.39	76.3%	0.0%
7	LN / W	Exp / LL	LL / LL	-222.38	76.3%	0.0%
8	W / W	LN / LL	LL / W	-222.37	71.1%	0.0%
9	LLog / W	LN / LL	LL / LL	-222.33	72.9%	0.0%
10	LN / W	LN / LL	LN / LL	-222.31	76.3%	0.0%

Distributions: LN = log-normal, W = weibull, LLog = log-logistic, Exp = exponential. C/P = Combo/Pembro. All top-ranked combinations use log-normal OS for the combo arm, yielding OS40c = 71.1–76.3% under unconstrained extrapolation, substantially higher than GP survival at age 101 ($\approx 0\%$). This motivated the GP mortality hazard floor (Section 5.1), under which the constrained OS40c falls to 0.0% for both arms.

4.3 Lifetime OS Extrapolation by Distribution

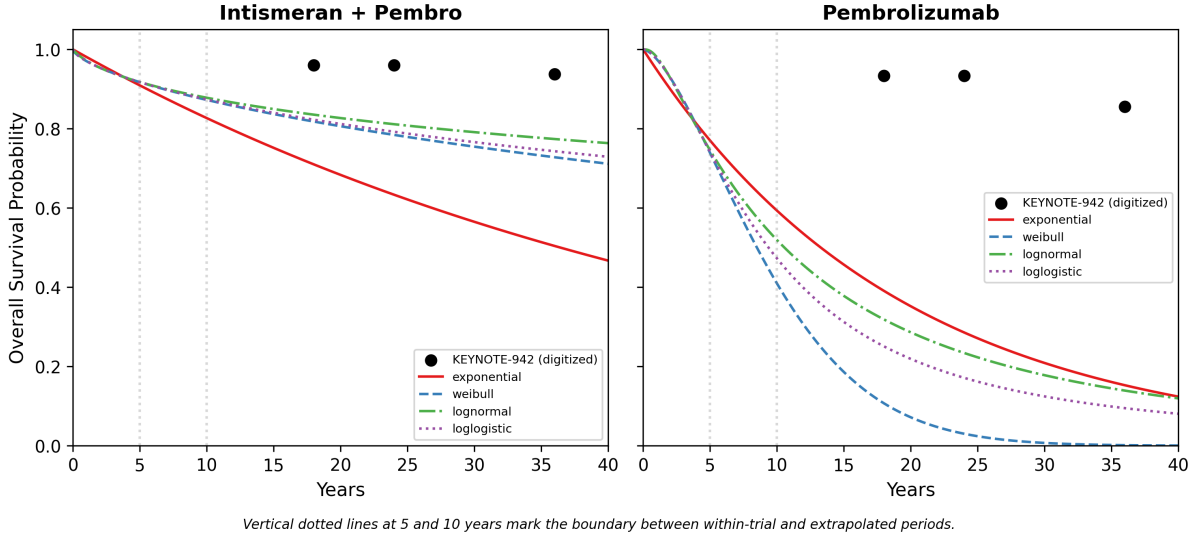


Figure S3: Lifetime OS extrapolation by parametric distribution. Within-trial data (0–5 years) are shown to the left of the first vertical dotted line; beyond 10 years is pure extrapolation. The four distributions produce wildly divergent 40-year survival estimates (0% to 76.3%).

5 General-Population Mortality and Structural Validation

5.1 GP Mortality Implementation

The GP mortality hazard floor is applied to modeled OS from month 60 onward.

Life table source: US 2019 period life tables (Arias et al., NVSS). **Baseline age:** 61 years (KEYNOTE-942 median). **Sex weighting:** 64% male, 36% female. **Interpolation:** Annual life table survival linearly interpolated to monthly resolution; monthly hazards derived as $h_{GP}(t) = -\ln[S_{GP}(t+1)/S_{GP}(t)]$.

Constraint formula (mortality hazard floor): from month 60 onward, the model mortality hazard is floored at the age- and sex-matched general-population hazard:

$$h_{\text{model}}^{\text{con}}(t) = \max[h_{\text{param}}(t), h_{\text{GP}}(t)] \quad t \geq 60 \text{ months}, \quad (1)$$

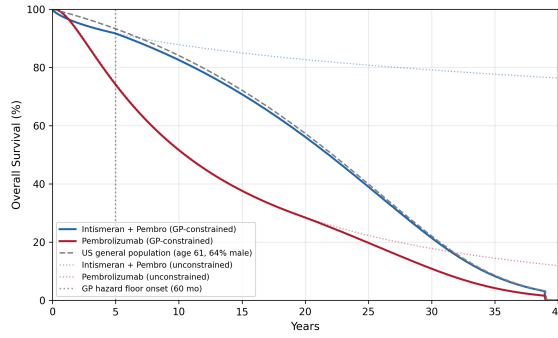
where $h_{\text{param}}(t) = -\ln[S_{\text{param}}(t+1)/S_{\text{param}}(t)]$ is the monthly hazard implied by the fitted parametric survival curve. The constrained survival is reconstructed as $S_{\text{model}}^{\text{con}}(t) = \exp[-\sum_0^t h^{\text{con}}(u)]$. This is a **hazard floor**, not a survival cap: a cohort that has already experienced elevated (cancer-related) mortality in the first 5 years remains below general-population survival at every subsequent time point, whereas a survival cap ($S = \min[S_{\text{param}}, S_{\text{GP}}]$) would incorrectly force the two curves to coincide whenever the parametric tail exceeded the general-population level. Both arms are subject to the same background mortality floor.

5.2 Long-Term Survival Diagnostics

Table S6: Long-Term Survival Predictions vs. General Population

Year	Combo OS	Pembro OS	GP Survival	Combo vs GP
5	91.6%	74.1%	93.3%	Below (excess mortality to date)
10	82.5%	51.6%	84.1%	Below (hazard floor active)
15	70.8%	37.6%	72.2%	Below (hazard floor active)
20	56.1%	28.4%	57.3%	Below (hazard floor active)
30	21.5%	10.9%	22.1%	Below (hazard floor active)
40	0.0%	0.0%	0.0%	At GP (age 101, GP survival ≈ 0)

5.3 OS versus General-Population Survival



GP-constrained curves (solid) use a mortality hazard floor from month 60. The constrained combo OS stays below the GP curve at all time points, reflecting the excess mortality experienced in the first 5 years. Unconstrained log-normal (dotted) produces implausibly high 40-year survival (76.4%).

Figure S4: OS versus general population. Solid lines: GP-constrained model (hazard floor from 60 months). Dotted lines: unconstrained log-normal extrapolation. Dashed gray line: US GP survival from age 61. The constrained combo curve remains below the GP curve at all time points, reflecting the hazard-floor property: a cohort with prior excess mortality does not "catch up" to general-population survival.

5.4 Structural Quality Assurance

Table S7: Structural Quality Assurance (480 Monthly Cycles)

Check	Result
Total cycles	480
RFS > DMFS violations	0
DMFS > OS violations	0
Negative RF state probability	0
Negative LR state probability	0
Negative DM state probability	0
Maximum state-sum error	$< 10^{-12}$
Clamp events (forced non-negative)	0

6 External Validation

6.1 External Validation Landmarks

Table S8: External Validation of Pembrolizumab Arm Predictions

Outcome	Year	Model (pembro)	KEYNOTE-054	CheckMate 238
RFS	5	40.5%	—	51% [6]
RFS	7	31.9%	50% [7]	46% [6]
DMFS	5	53.8%	—	59% [6]
DMFS	7	44.9%	54% [7]	55% [6]
OS	5	74.5%	NR	76% [8]
OS	7	64.2%	NR	71% [6]

NR: Not reported. CheckMate 238 evaluated nivolumab (not pembrolizumab). The model predicts lower RFS/DMFS versus external trials, reflecting the higher-risk KEYNOTE-942 population (stage IIIB–IV including resected stage IV melanoma vs IIIA–C in KEYNOTE-054 and IIIB–IV in CheckMate 238). Only published landmark values are included; no extrapolated estimates are used.[6, 7]

6.2 Longitudinal External Validation

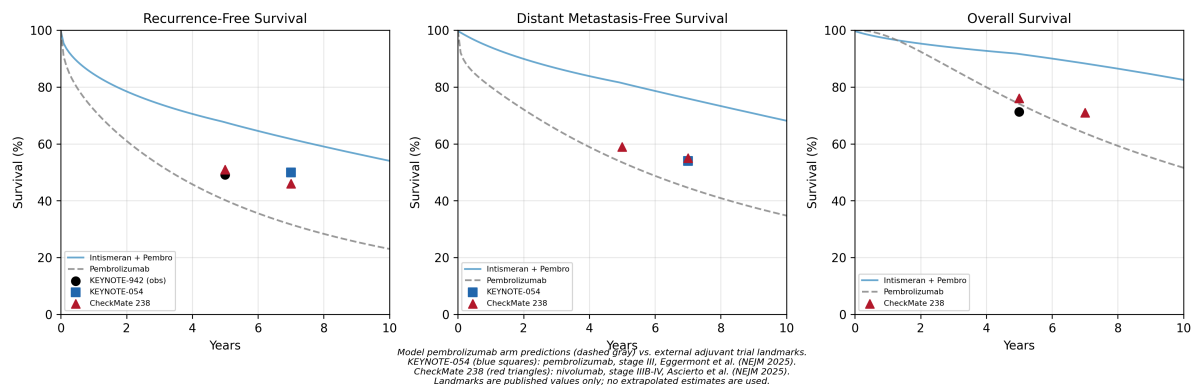


Figure S5: External validation of pembrolizumab arm predictions. Gray line: model. Circles: KEYNOTE-942 observed data. Squares: KEYNOTE-054 (pembrolizumab, 7-year). Triangles: CheckMate 238 (nivolumab, 5- and 7-year). Only published landmark values are shown.

7 Model Inputs

7.1 Complete Parameter Table

Table S9: Full Model Parameter Inventory

Parameter	Base	DSA Low	DSA High	PSA Distribution	Source
<i>Clinical Parameters</i>					
OS μ – combo	8.268	4.228	4.628	Normal (SE from Jacobian)	Fitted
OS σ – combo	0.478	0.378	0.578	Normal (SE from Jacobian)	Fitted
OS μ – pembro	3.725	3.525	3.925	Normal (SE from Jacobian)	Fitted
OS σ – pembro	0.574	0.474	0.674	Normal (SE from Jacobian)	Fitted
r _{DM} μ – combo	4.107	3.907	4.307	Normal (SE from Jacobian)	Fitted
r _{DM} σ – combo	0.432	0.332	0.532	Normal (SE from Jacobian)	Fitted
r _{DM} μ – pembro	2.849	2.649	3.049	Normal (SE from Jacobian)	Fitted
r _{DM} σ – pembro	0.934	0.834	1.034	Normal (SE from Jacobian)	Fitted
r _{LR} μ – combo	3.367	3.167	3.567	Normal (SE from Jacobian)	Fitted
r _{LR} σ – combo	0.624	0.524	0.724	Normal (SE from Jacobian)	Fitted
r _{LR} μ – pembro	2.750	2.550	2.950	Normal (SE from Jacobian)	Fitted
r _{LR} σ – pembro	0.638	0.538	0.738	Normal (SE from Jacobian)	Fitted
Treatment duration (pembro)	18 cycles (2 yr)	9	24	Fixed	[4]
Treatment duration (intismeran)	9 doses (1 yr)	6	12	Fixed	[4]
Baseline age	61 years	55	67	Fixed	[4]
Male proportion	64%	–	–	Fixed	[4]
<i>Costs (2026 USD)</i>					
Intismeran (one-time)	\$200,000	\$160,000	\$240,000	Gamma (CV=0.2)	Assumed
Pembrolizumab (annual)	\$220,896	\$176,717	\$265,075	Gamma (CV=0.2)	WAC (CMS)
Tumor sequencing	\$1,000	\$500	\$2,000	Gamma (CV=0.2)	CMS
Administration (per cycle)	\$500	\$250	\$750	Gamma (CV=0.2)	CMS
AE management (one-time)	\$15,000	\$7,500	\$22,500	Gamma (CV=0.2)	[3]
LR management (monthly)	\$3,000	\$2,400	\$3,600	Gamma (CV=0.2)	[3]
DM management (monthly)	\$12,000	\$9,600	\$14,400	Gamma (CV=0.2)	[1]
<i>Utilities</i>					
RF state	0.83	0.80	0.86	Beta (SE=0.03)	[2]
LR state	0.64	0.61	0.67	Beta (SE=0.03)	[2]
DM state	0.55	0.52	0.58	Beta (SE=0.03)	[2]
AE disutility (one-time)	0.05	0.00	0.10	Beta (SE=0.02)	Assumed
<i>Economic Parameters</i>					
Discount rate (costs, effects)	3%	0%	5%	Fixed (DSA only)	[9]
Time horizon	40 years	10	20	Fixed (scenario)	Assumed
Willingness-to-pay threshold	\$100K–\$150K/QALY	\$50K	\$200K	Fixed (scenario)	[10]
Cost base year	2026 USD	–	–	–	Assumed
GP mortality source	US 2019 Life Tables	–	–	–	NVSS (Arias 2022)

8 Additional Results

8.1 Lifetime Cost Decomposition

Table S10: Lifetime Cost Decomposition by Component

Component	Combo	Pembro	Incremental
Intismeran	\$200,000	\$0	\$200,000
Pembrolizumab	\$217,888	\$217,888	\$0
Sequencing	\$1,000	\$0	\$1,000
Administration	\$493	\$493	\$0
Adverse events	\$15,000	\$0	\$15,000
LR management	\$112,203	\$74,477	\$37,726
DM management	\$524,121	\$453,082	\$71,039
Total	\$1,070,705	\$745,939	\$324,765

The combination arm incurred \$71,039 higher lifetime DM management costs, reflecting longer survival and greater cumulative exposure to downstream disease-management costs.

8.2 LY and QALY Decomposition

Table S11: Life-Year and QALY Decomposition by Health State

Outcome	Combo	Pembro	Incremental
Life-years (undiscounted)	23.78	15.34	7.13
Life-years (discounted 3%)	19.01	10.75	8.26
RF state QALYs	10.17	4.60	5.57
LR state QALYs	1.99	1.32	0.67
DM state QALYs	2.00	1.73	-0.32
Total QALYs	14.16	7.65	6.52

Health benefit is driven by RF state QALYs (5.57 gained), more than compensating for fewer DM-state QALYs (-0.32).

9 Uncertainty and Structural Scenario Analyses

9.1 Deterministic Sensitivity Analysis

Table S12: One-Way Deterministic Sensitivity Analysis

Parameter	Base	Low	High	ICER range	Change in NMB
Utility RF	0.83	0.80	0.86	51,431–48,346	\$60,441
Utility LR	0.64	0.61	0.67	50,082–49,601	\$9,431
Utility DM	0.55	0.52	0.58	49,954–49,728	\$4,440
Pembrolizumab annual cost	\$220,896	\$176,717	\$265,075	49,841–49,841	\$0
Intismeran cost	\$200,000	\$160,000	\$240,000	43,702–55,979	\$80,000
LR monthly cost	\$3,000	\$2,400	\$3,600	48,683–50,999	\$15,090
DM monthly cost	\$12,000	\$9,600	\$14,400	47,660–52,021	\$28,416
Discount rate	3%	0%	5%	49,330–53,363	\$798,812
OS μ combo	8.396	5.037	11.754	51,465–49,690	\$627,624
OS μ pembro	4.841	3.873	5.810	60,695–Dominant	\$494,048

The ICER remained below \$61,000/QALY across all tested ranges, with the combination dominant (lower cost and higher QALY) when the pembrolizumab OS parameter was increased by 20%.

9.2 Probabilistic Sensitivity Analysis

9.2.1 PSA Implementation Details

PSA with 6,000 Monte Carlo iterations (seed = 42). Survival parameters (μ , σ for 6 curves) drawn jointly from bivariate normal via Jacobian covariance ($r > 0.95$). Costs: gamma (CV=0.2). Utilities: beta (SE=0.03). Structural constraints (RFS $\leq$ DMFS $\leq$ OS, monotonicity, survival within [0,1]) verified every cycle; all 6,000 draws were structurally valid (0 rejections). No iterations were excluded on the basis of the ICER value.

Table S13: PSA Parameter Distributions and Sampled Values

Parameter	Distribution	Fitted mean	SE	Correlation	Sampling
OS μ / σ – combo	Bivariate normal	8.396 / 3.096	0.793 / 0.522	0.996	Joint
OS μ / σ – pembro	Bivariate normal	4.841 / 1.131	0.291 / 0.310	0.955	Joint
r _{DM} μ / σ – combo	Bivariate normal	7.113 / 2.480	0.814 / 0.563	0.995	Joint
r _{DM} μ / σ – pembro	Bivariate normal	7.042 / 5.000	2.278 / 3.130	0.995	Joint
r _{LR} μ / σ – combo	Bivariate normal	8.874 / 5.000	2.039 / 1.928	0.997	Joint
r _{LR} μ / σ – pembro	Bivariate normal	5.913 / 2.674	0.757 / 0.879	0.984	Joint
Intismeran cost	Gamma (CV=0.2)	\$200,000	—	—	Independent
Pembrolizumab annual cost	Gamma (CV=0.2)	\$220,896	—	—	Independent
LR monthly cost	Gamma (CV=0.2)	\$3,000	—	—	Independent
DM monthly cost	Gamma (CV=0.2)	\$12,000	—	—	Independent
Utility RF / LR / DM	Beta (SE=0.03)	0.83 / 0.64 / 0.55	—	—	Independent

9.2.2 PSA Results

- Mean Δ QALY: 6.65 (95% CI 4.13–9.83).
- Mean Δ Cost: \$293,238 (95% CI –\$533,282 to \$825,708).
- Mean NMB: \$371,570 at \$100K and \$703,975 at \$150K.
- Probability cost-effective: 52.4% at \$50K, 93.4% at \$100K, and 98.4% at \$150K.
- No iterations were excluded based on the ICER; 11.2% of draws were cost-saving, and all 6,000 draws had Δ QALY $\geq$ 0.

9.2.3 PSA Convergence

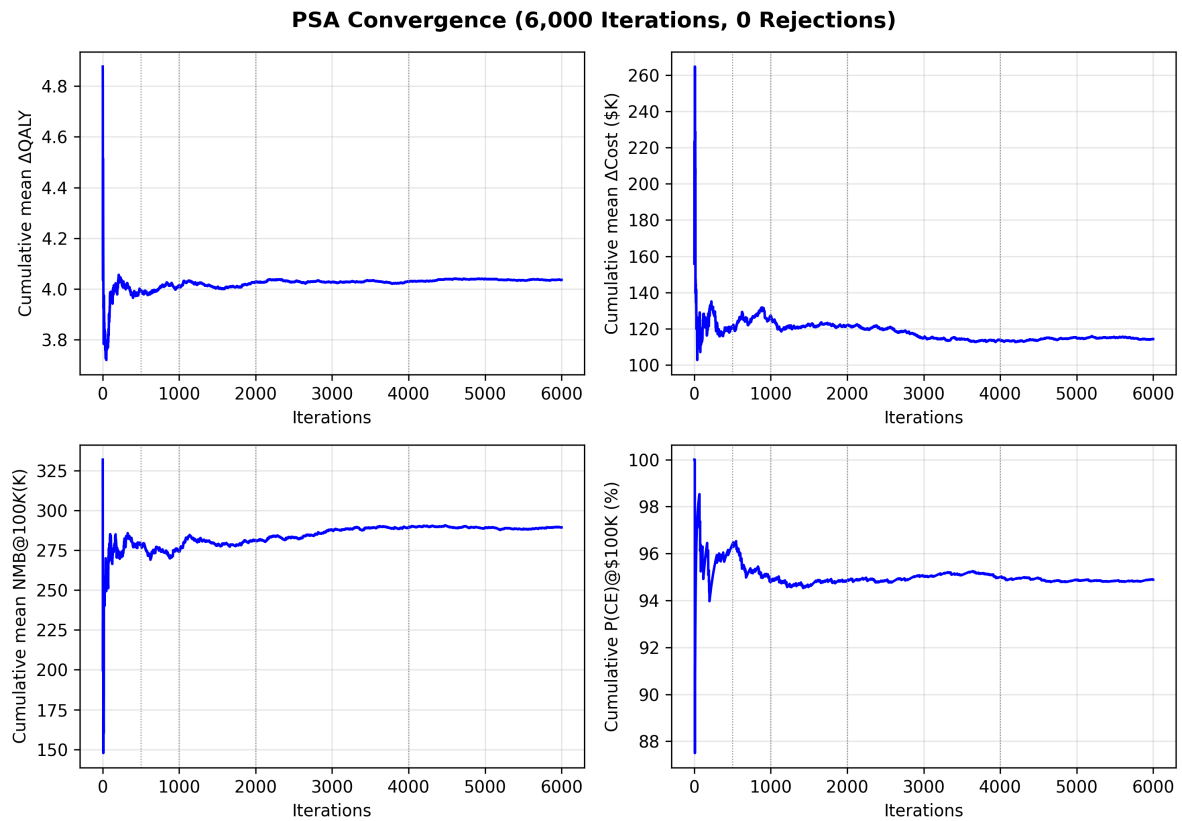


Figure S6: PSA convergence across iteration counts (6,000 structurally valid iterations, 0 structural rejections, seed = 42). Cumulative means of Δ QALY, Δ Cost, NMB, and P(CE). Vertical dotted lines at 500, 1,000, 2,000, and 4,000 iterations. Estimates stable from $\sim$ 500 iterations.

9.2.4 Cost-Effectiveness Plane

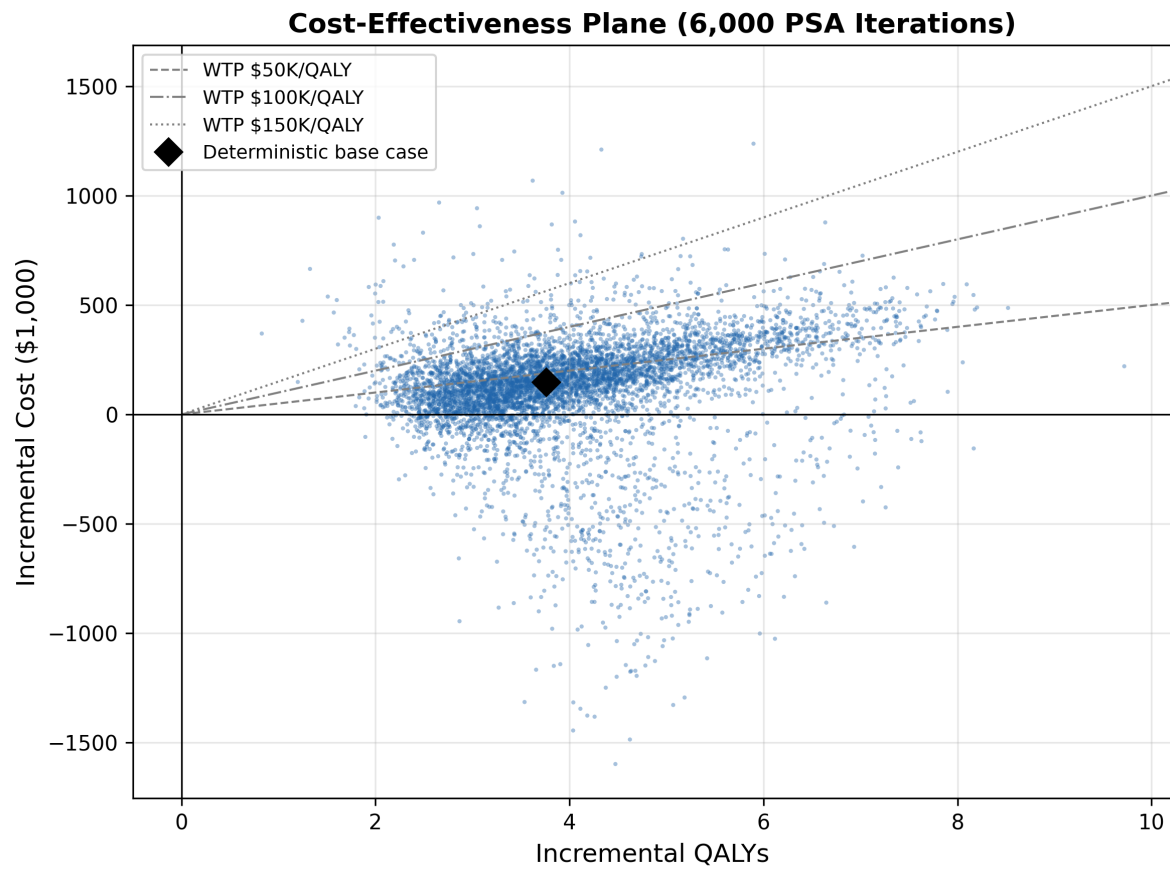


Figure S7: CE plane showing 6,000 PSA iterations (blue), WTP lines (\$50K, \$100K, \$150K per QALY), and the deterministic base case (black diamond). 11.2% of iterations fall in the southeast quadrant (cost-saving); all 6,000 iterations had positive ΔQALY . No iterations were excluded based on the ICER value.

9.3 Structural Scenario Analyses

Table S14: Structural Scenario Analysis Results

Scenario	ΔLY	$\Delta QALY$	$\Delta Cost$	ICER	NMB \$100K	NMB \$150K
Base case	8.26	6.52	\$324,765	\$49,841	\$326,840	\$652,643
Weibull OS	11.05	8.40	\$503,437	\$59,959	\$336,193	\$756,008
Unconstrained (no GP)	9.24	7.22	\$373,488	\$51,753	\$348,189	\$709,028
GP floor 48 mo	8.26	6.52	\$324,765	\$49,841	\$326,840	\$652,643
GP floor 60 mo (base)	8.26	6.52	\$324,765	\$49,841	\$326,840	\$652,643
GP floor 72 mo	8.26	6.52	\$324,765	\$49,841	\$326,840	\$652,643
GP floor 96 mo	8.28	6.53	\$325,801	\$49,862	\$327,608	\$654,313
10-year horizon	1.35	1.34	\$97,868	\$73,233	\$35,771	\$102,591
20-year horizon	4.15	3.52	\$147,215	\$41,785	\$205,099	\$381,256
30-year horizon	6.54	5.29	\$239,745	\$45,349	\$288,922	\$553,256
40-year horizon (base)	8.26	6.52	\$324,765	\$49,841	\$326,840	\$652,643
Intismeran \$100K	8.26	6.52	\$224,765	\$34,494	\$426,840	\$752,643
Intismeran \$200K (base)	8.26	6.52	\$324,765	\$49,841	\$326,840	\$652,643
Intismeran \$300K	8.26	6.52	\$424,765	\$65,187	\$226,840	\$552,643
Intismeran \$500K	8.26	6.52	\$624,765	\$95,881	\$26,840	\$352,643
Waning V1 (recurrence)	2.09	2.09	\$39,862	\$19,055	\$169,330	\$273,926
Waning V2 (hazard convergence)	3.05	2.39	\$249,355	\$104,247	-\$10,159	\$109,439
No direct OS benefit	0.00	0.52	-\$22,543	Dominant	\$74,262	\$100,121

The base-case conclusion is robust across structural assumptions. The ICER alone is insufficient: unconstrained model has moderate ICER (\$51,753) but implausible \$373,488 cost.

Time horizon sensitivity. The ICER remained below conventional US willingness-to-pay thresholds across 20-, 30-, and 40-year horizons, although the point estimate increased at the 40-year horizon: \$41,785 at 20 years, \$45,349 at 30 years, and \$49,841 at the 40-year base-case horizon (Table S14). The 30-to-40-year increase of \$4,492/QALY (9.9%) reflects the contribution of late discounted QALY gains that are not fully captured at shorter horizons; the 10-year horizon produced a notably higher ICER of \$73,233/QALY, reflecting the incomplete capture of long-term QALY gains within a shorter window. Because the general-population mortality hazard floor was applied from 60 months onward and both costs and QALYs were discounted at 3% annually, survival beyond the 20-year window contributed materially to discounted incremental costs and QALYs.

General population hazard floor start-time sensitivity. The GP mortality floor was applied from 60 months in the base case. Varying the onset from 48 to 96 months produced no material change in the ICER (all \$49,841/QALY when rounded to whole dollars; the 96-month onset yielded \$49,862/QALY). This reflects the fact that the fitted hazard for the combination arm closely tracks the general-population mortality hazard in the post-trial period (both at approximately 0.0008–0.0009 monthly), leaving the floor with minimal additional constraining effect regardless of onset timing. The floor’s binding is thus a near-numerical coincidence between the parametric extrapolation and US life-table mortality, not a model artifact.

10 Deterministic Threshold Acquisition Price

10.1 Price Threshold Analysis

Table S15: Price Threshold Analysis

Intismeran Price	ICER	NMB @ \$100K	NMB @ \$150K
\$0	\$19,147	\$526,835	\$852,635
\$50,000	\$26,821	\$476,835	\$802,635
\$100,000	\$34,494	\$426,835	\$752,635
\$150,000	\$42,168	\$376,835	\$702,635
\$200,000 (base)	\$49,841	\$326,835	\$652,635
\$250,000	\$57,515	\$276,835	\$602,635
\$300,000	\$65,188	\$226,835	\$552,635
\$350,000	\$72,861	\$176,835	\$502,635
\$400,000	\$80,535	\$126,835	\$452,635
\$450,000	\$88,208	\$76,835	\$402,635
\$500,000	\$95,882	\$26,835	\$352,635
\$526,840	\$100,000	\$0	\$300,000
\$550,000	\$103,555	-\$23,165	\$302,635
\$600,000	\$111,229	-\$73,165	\$252,635
\$650,000	\$118,902	-\$123,165	\$202,635
\$700,000	\$126,575	-\$173,165	\$152,635
\$750,000	\$134,249	-\$223,165	\$102,635
\$800,000	\$141,922	-\$273,165	\$52,635
\$852,643	\$150,000	-\$325,000	\$0

Deterministic threshold prices (from Table S15): approximately \$526,840 at \$100,000/QALY (2.6× base) and approximately \$852,643 at \$150,000/QALY (4.3× base). These are model-based deterministic estimates; PSA uncertainty propagation and formal budget-impact analysis would be needed to inform payer negotiations.

10.2 ICER versus Acquisition Price

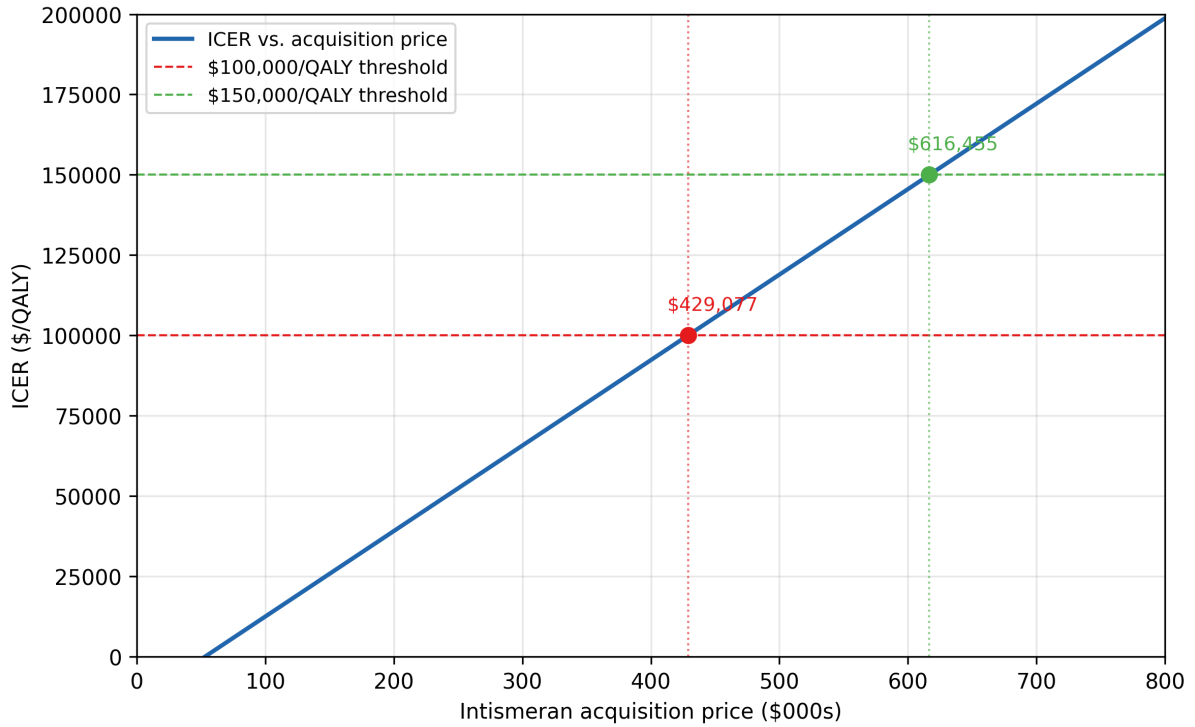


Figure S8: ICER as a function of intismeran acquisition price. Red/green dashed lines: \$100K/\$150K thresholds. Vertical dotted lines: threshold prices.

11 Reproducibility and Reporting

11.1 Technical Reproducibility

All analyses were performed using Python 3.12.7 (Apple Clang 15.0.0) on macOS 26.6.2 (Apple Silicon). The full code repository is available at <https://github.com/yichao2022/cea-intismeran> (tag v1.0-submission).

Key dependencies: NumPy 1.26.4, SciPy 1.14.1, Matplotlib 3.9.3, Pandas 2.2.3, LaTeX (TeX Live 2025, pdfLaTeX + BibTeX + natbib).

Model details:

- Cycle length: monthly (1/12 year); time horizon: 40 years (480 cycles)
- Half-cycle correction: not applied (monthly cycles are sufficiently fine-grained)
- Discounting: continuous-time exponential: $1/(1+r)^t$ where $r = 0.03$
- Random seed: 42 (PSA reproducibility)
- GP mortality source: US 2019 period life tables (NVSS, Arias et al. 2022)

Code structure:

- `model.py` — ModelParams, CEAModel (4-state PSM)
- `rerun_primary.py` — CEAModelV2 (hazard-based GP floor, waning, no-OS)
- `psa_full.py` — PSA with joint survival sampling
- `survival_model_selection.py` — 4,096-combination model selection
- `final_run.py` / `threshold_analysis.py` — DSA, scenarios, pricing
- `figures.py` / `gen_fig*.py` — Figure generation scripts

- `output/survival_combos_4096.csv` — Full model selection results

Data extraction: Survival probabilities extracted from Figure 1 of Khattak et al. (2026) using WebPlotDigitizer v4.7. The digitized values are in Table S2.

Reproducibility: Running `python3 rerun_primary.py` reproduces base-case results. `python3 psa_full.py` reproduces PSA. GitHub tag `v1.0-submission` freezes the submission code.

11.2 CHEERS 2022 Checklist

Table S16: CHEERS 2022 Checklist

No.	Item	Reported Location	
1	Title	Yes	Title page
2	Abstract	Yes	Abstract
3	Background	Yes	Introduction
4	Health economic analysis plan	No	This is a retrospective analysis, not a pre-specified HEAP
5	Model	Yes	Methods, Appendix A
6	Population	Yes	Methods
7	Setting	Yes	Methods
8	Comparators	Yes	Methods
9	Perspective	Yes	Methods
10	Time horizon	Yes	Methods
11	Discount rate	Yes	Methods
12	Outcomes	Yes	Methods
13	Health states	Yes	Appendix A
14	Measurement of outcomes	Yes	Methods
15	Resource use	Yes	Methods
16	Currency	Yes	Methods
17	Model choice	Yes	Methods
18	Assumptions	Yes	Methods (§2.2–2.7)
19	Analytic methods	Yes	Methods (§2.7)
20	Study parameters	Yes	Methods (§2.4, 2.5), Appendix D
21	Incremental results	Yes	Results (§3.1–3.2)
22	Uncertainty	Yes	Results (§3.4–3.6), Supplement
23	Heterogeneity	Not assessed	Limitations (§4.7, subgroup heterogeneity acknowledged)
24	Discussion	Yes	Discussion
25	Limitations	Yes	Discussion
26	Funding	Yes	Declarations
27	Conflicts	Yes	Declarations
28	Reproducibility	Yes	Appendix

References

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